Computer Science | Data Science | Moscow 2026

Network Science: Homework Project

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Code link： https://github.com/HuJinnan/NetworkScience_Projiect_HuJinnan

1. Data choice
2. Network construction
3. Analysis
4. Conclusion

I use my "Weibo" account to build my **social network**, because it is widely used in China, and it is the only app which everyone can see other user's follower list.

1. First, I use a program to find and record all of **my followings** (friends).

2. Second, I use my friend's ID to find **their followings**.

3.The following list located in website "**https://weibo.com/u/page/follow/One's_ID**". If one's followings list is too large, others can only see **no more than 220** of them

I have 190 followings, and they have more than 23K followings in total.

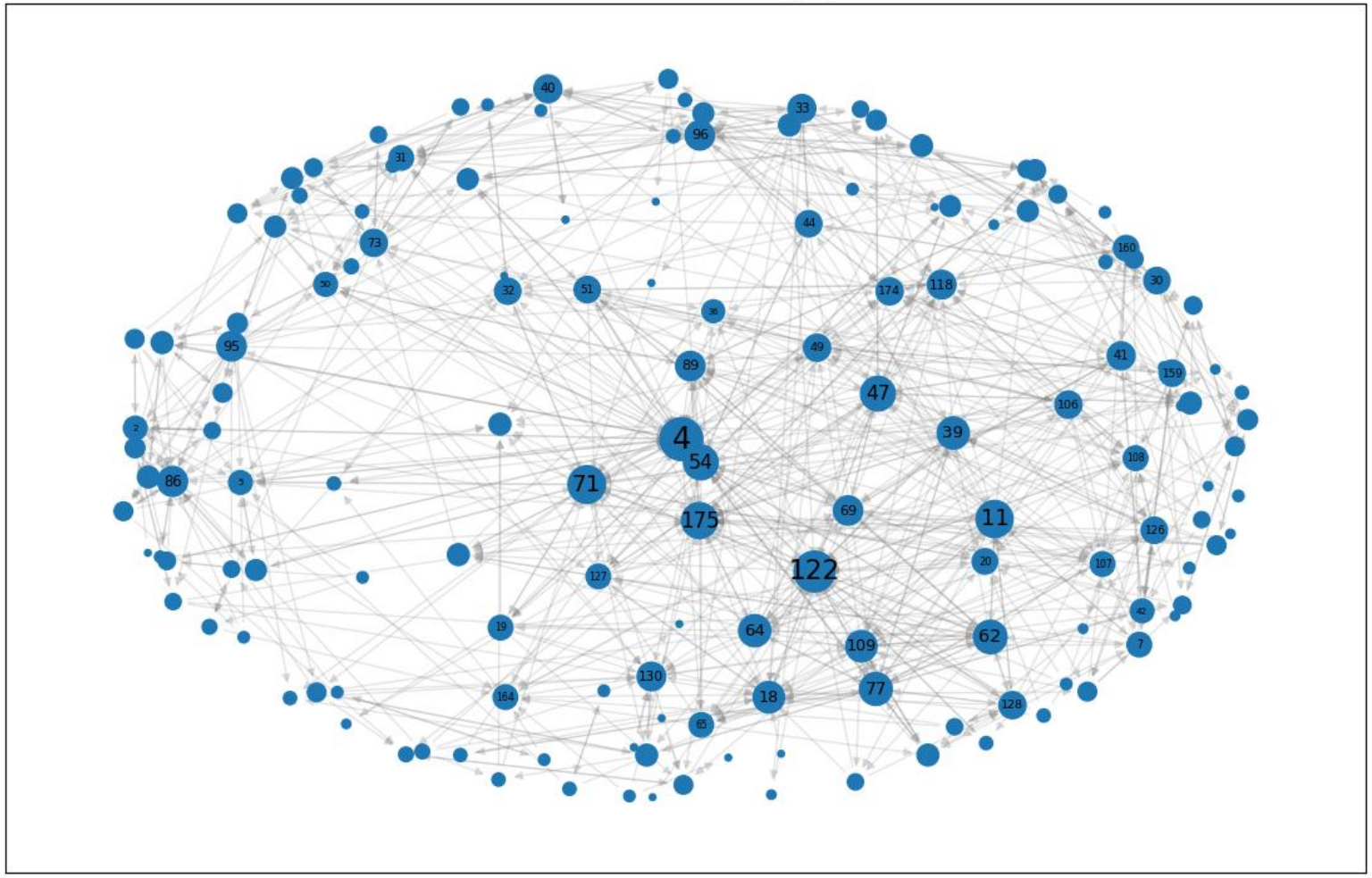
```
My friends list information is :-----
          uid          name
0  7834343861     直言trigger
1  2163365571  3DMGAME官方微博
2  1947605001       糖七又是糖九
3  1843450093           关大洲
4  6603744955      Yocar-冯骥
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 190 entries, 0 to 189
```

```
My friend's following list information is :--------------------
   friend_uid friend_name  follow_uid  follow_name
0  7834343861   直言trigger  6603744955     Yocar-冯骥
1  7834343861   直言trigger  2998395293  Astell_Kern
2  7834343861   直言trigger  7305759370       肉扣热热子
3  7834343861   直言trigger  1985698157     SVIP内容精选
4  7834343861   直言trigger  7579562933      微博社交会员
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23260 entries, 0 to 23259
```

I just want to build the relationship related to myself, so I only keep the **relationship between my followings**. After the cleaning, there are only 761 relationship left.

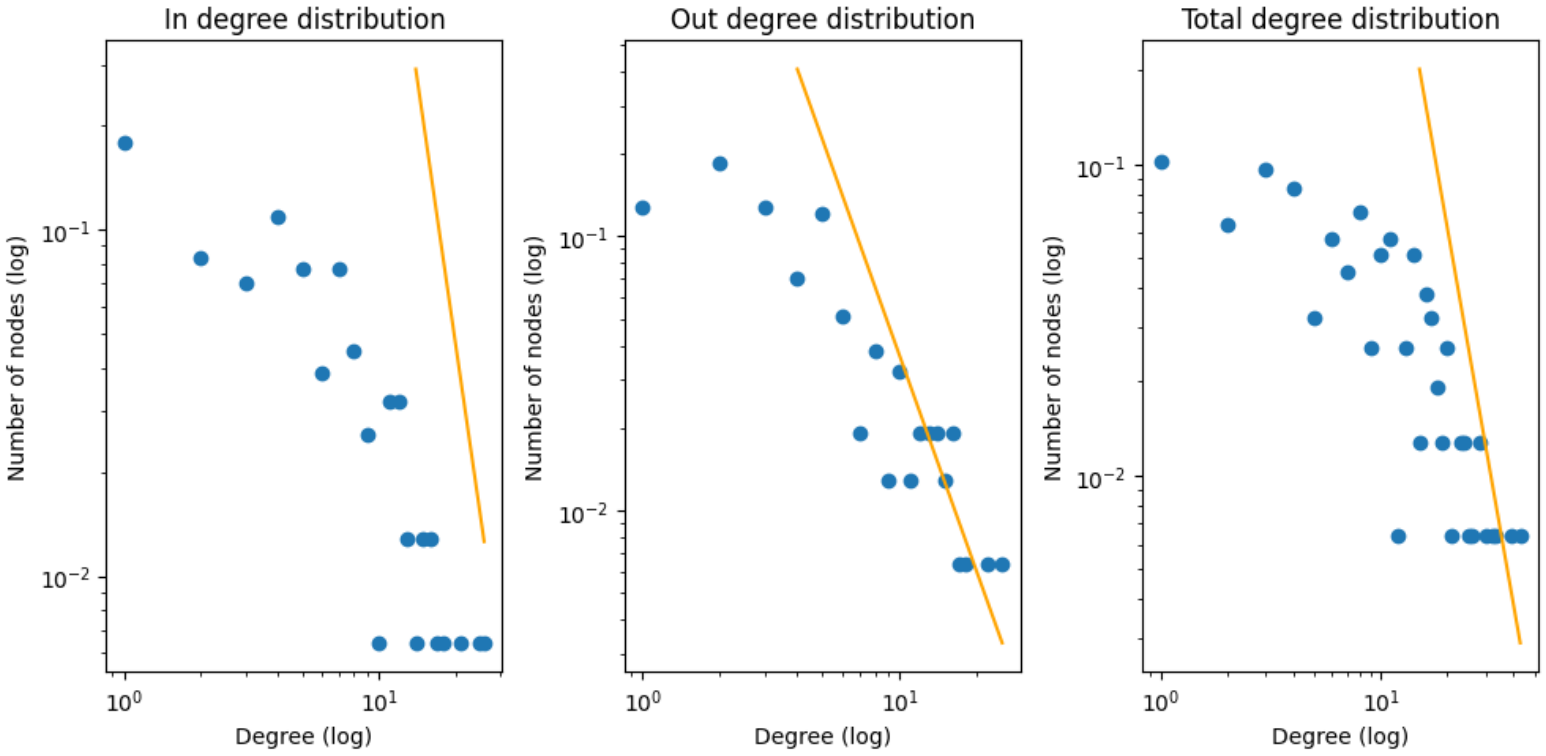
After removing the independent nodes, my social network ultimately became an unweighted network with 156 nodes and 761 directed edges.

Connected friend relation graph



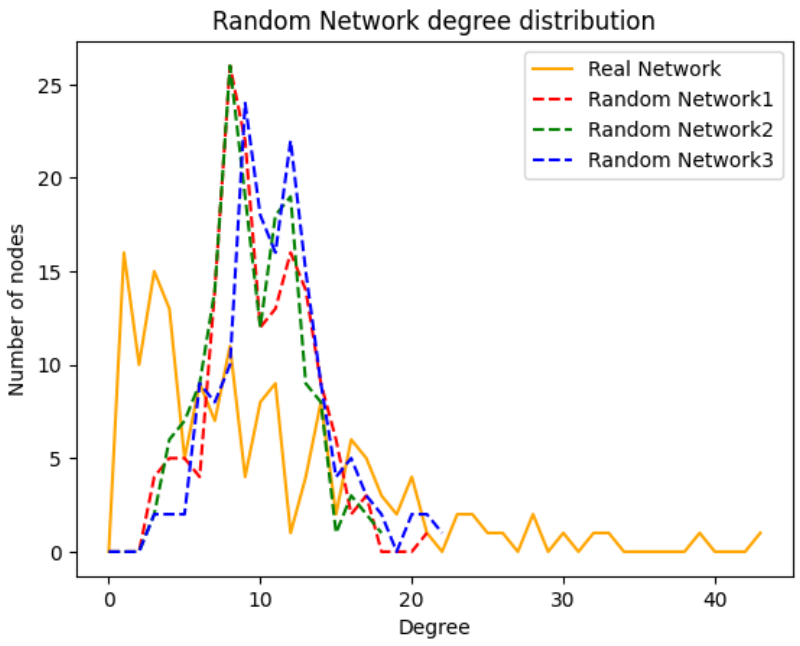
I think the out and total degree of this network are close to a power-law distribution in the high degree range. However, because the total number of points is not large, so the fitting distance become larger in the interval of lower degrees.

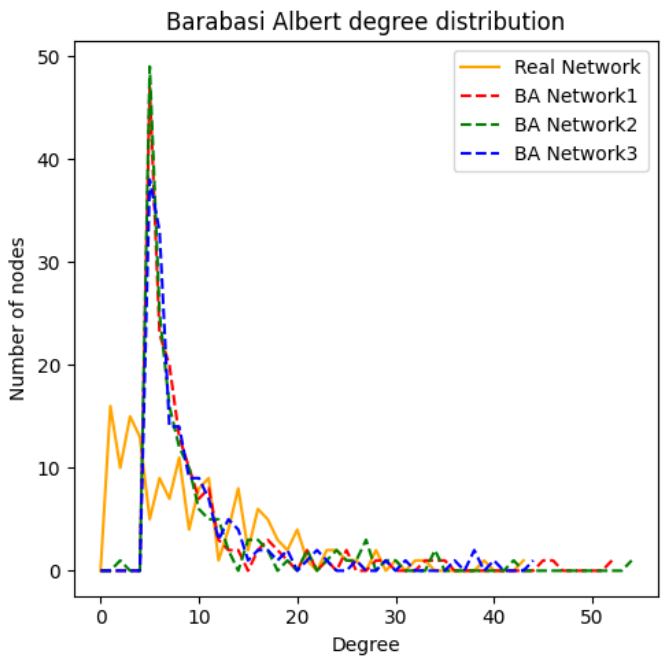
I also calculated assortativity and the result was -0.02974. This indicates that network exhibits a slight disassortativity, meaning that nodes with high degrees tend to connect to those with low degrees. This is **consistent with the power-law distribution** pattern analyzed earlier.

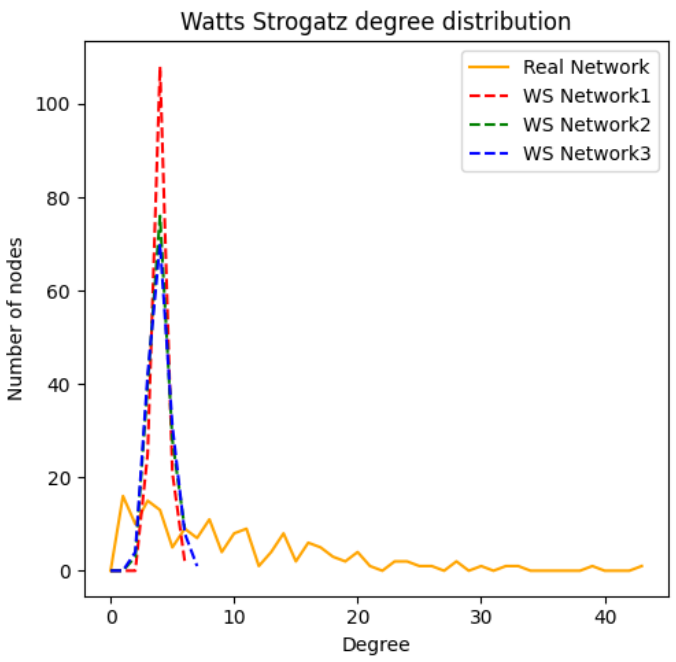


It is obvious that the Barabasi Albert model can fit the real network well in the **range of larger degrees**, but it has a significantly higher distribution probability when the degree is small.

This is consistent with the results of our **previous analysis of the power-law distribution** fitting of the real model, because the real network has fewer nodes, thus there is a larger gap in the region of smaller degrees.







We can observe that the average shortest path length of the real network is 3.226 and the clustering coefficient is 0.1936. Compared with the random network, its average shortest path length is smaller and the clustering coefficient is higher. We can conclude that the real network conforms to the small-world property.

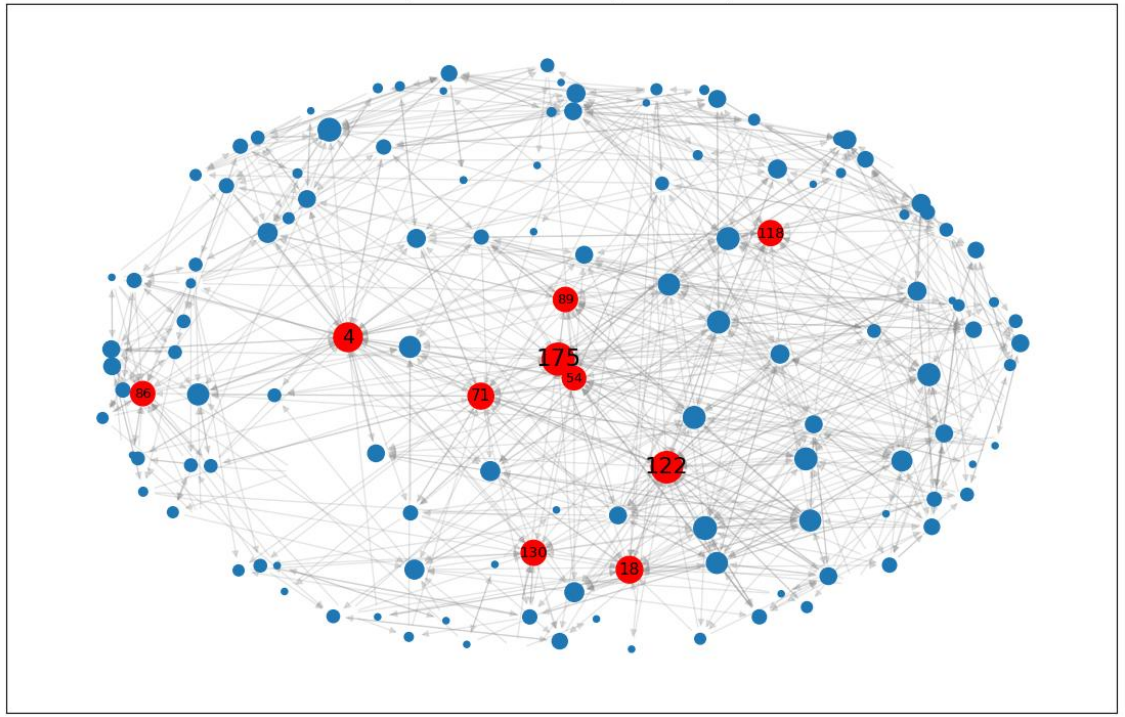
The Watts Strogatz model is clearly inconsistent with real networks, and all the results are significantly higher than those of real networks. Therefore, it is completely **unable to replicate the original structure** of that network.

```
For real network:-----------------------------
The average shortest path length is: 3.266
The clustering coefficient is: 0.1936

For all network model:------------------------------
The average shortest path length are: ER:3.341, BA:3.149, WS:8.237
The clustering coefficient are: ER:0.0611, BA:0.1444, WS:0.3721

All distance are:-----------------------------
The average shortest path length distance are: ER:-0.07584, BA:0.117, WS:-4.972
The clustering coefficient distance are: ER:0.1325, BA:0.04919, WS:-0.1786
```

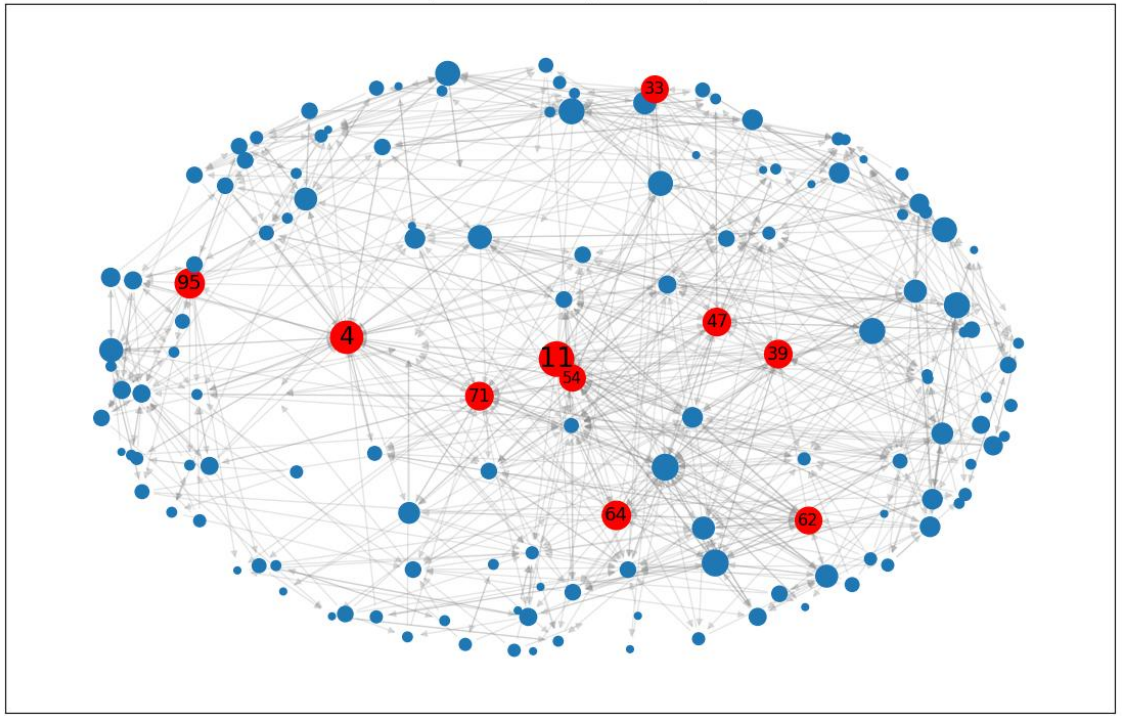
I found the top 10 nodes using different methods, and the results are as follows.

We can see that for degree centrality, the **common elements** of hubs and authorities are only three **[4, 54,71].** However, the overall node sizes presented in the images are **very similar**, indicating that the differences between the two are not very significant.

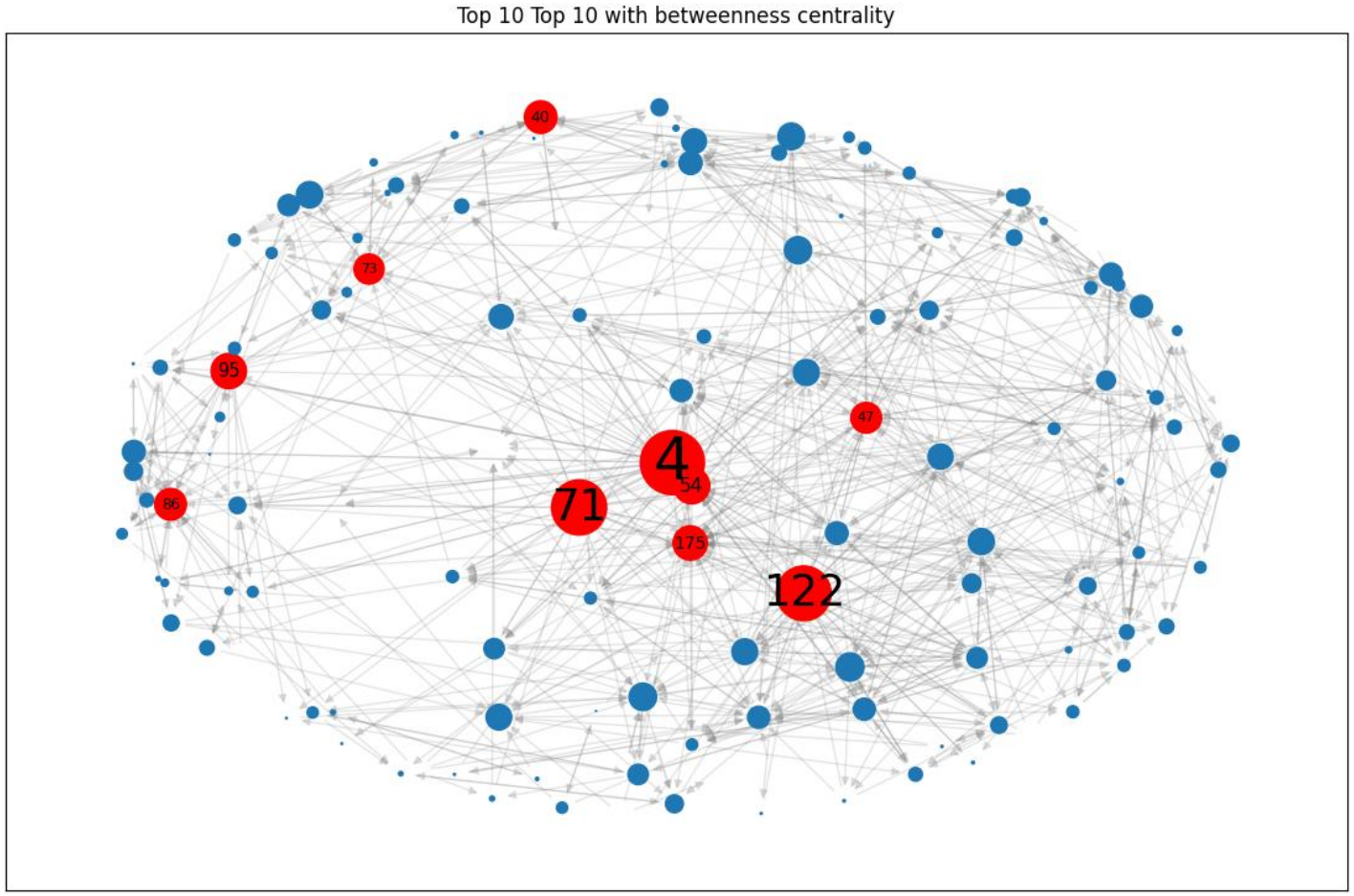
Top 10 Authorities with degree centrality



Top 10 Hubs with degree centrality

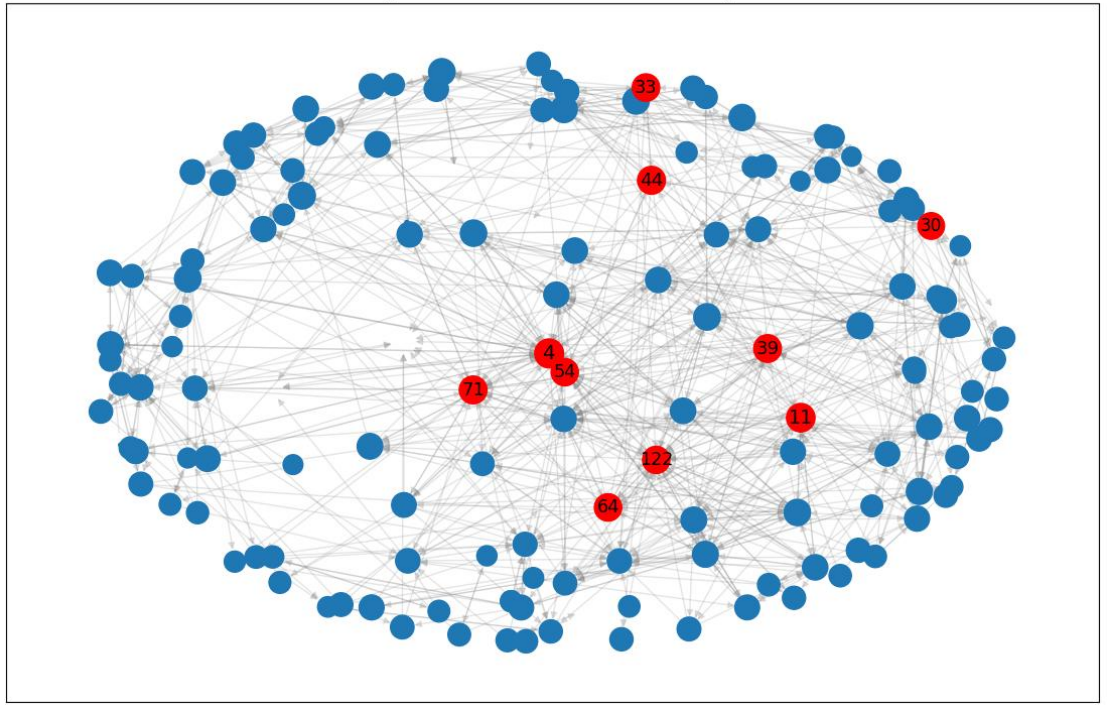


The gap between the top 10 nodes calculated by betweenness centrality is still quite significant. Among them, **[4, 71, 122]** are the three **largest**, and they are approximately **twice as large as** the remaining top 10.

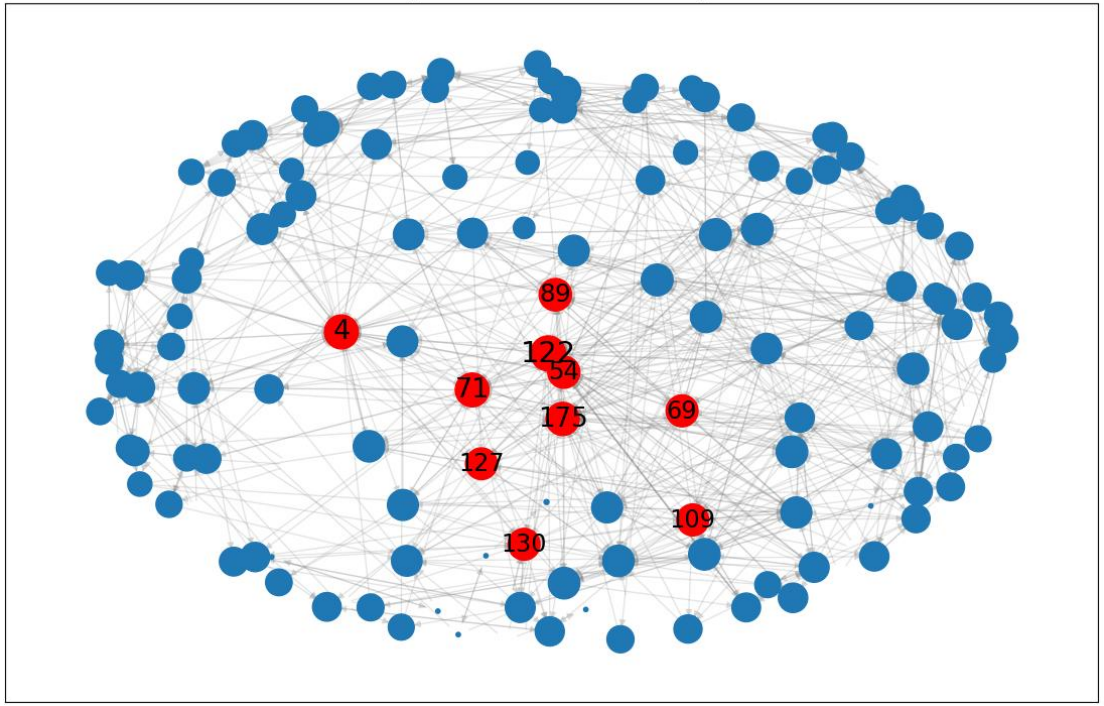


When using closeness centrality, the sizes of most nodes are **relatively similar**, which indicates that my network **has small-world characteristics**. The average shortest path length from all nodes to other nodes is uniform, and there are no extremely remote nodes. This is consistent with the previous analysis conclusion.
The common elements of Hubs and authorities are: **[122, 4, 54, 71].**

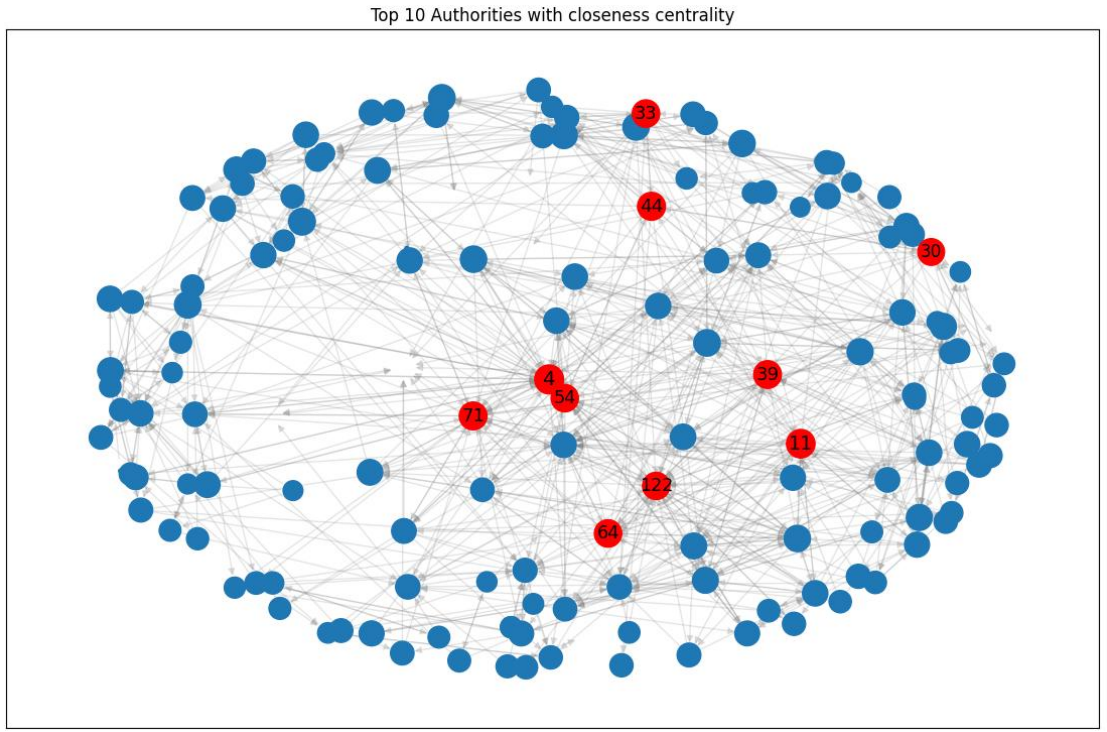
Top 10 Authorities with closeness centrality

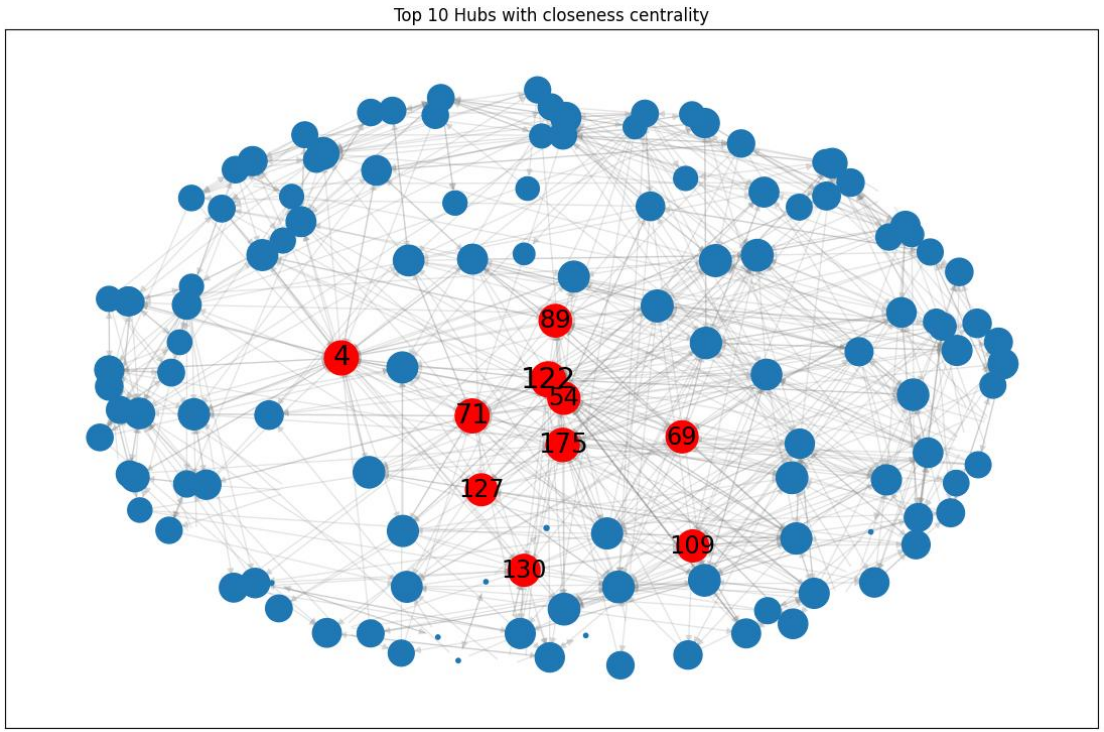


Top 10 Hubs with closeness centrality



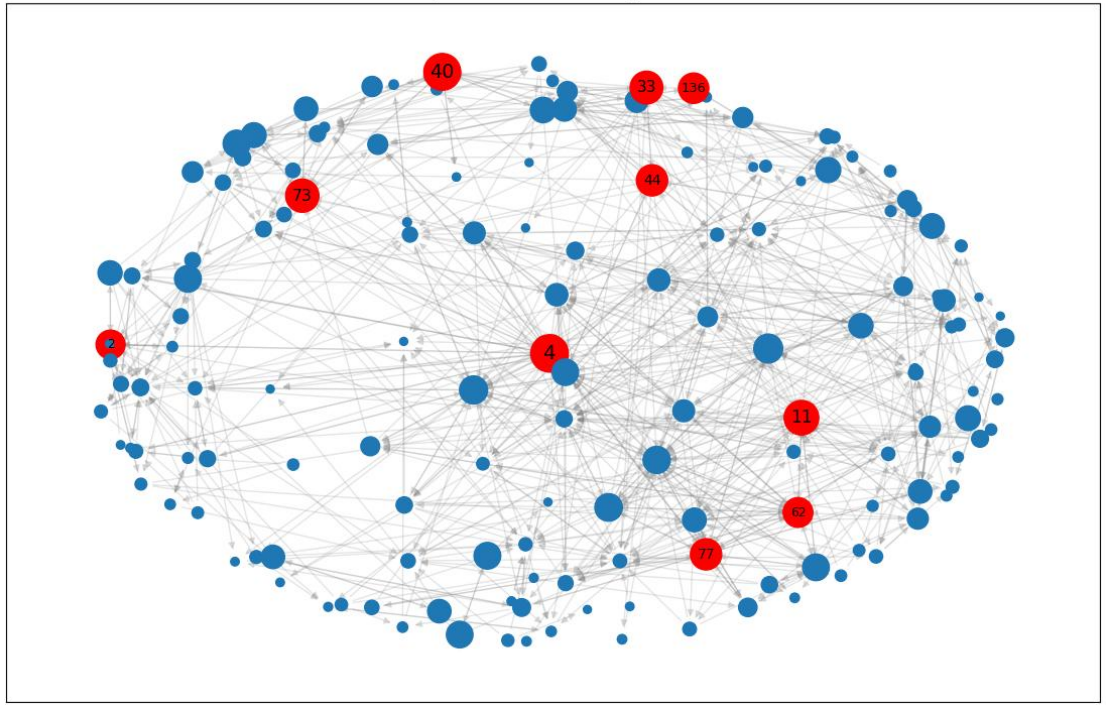
After calculating the eigenvector centrality, we can see that when both "in" and "out" are used, **five** common top 10 items are found: **[4, 77, 54, 122, 62].** They have indirect influence because they are connected to **neighbor nodes with significant influence**.



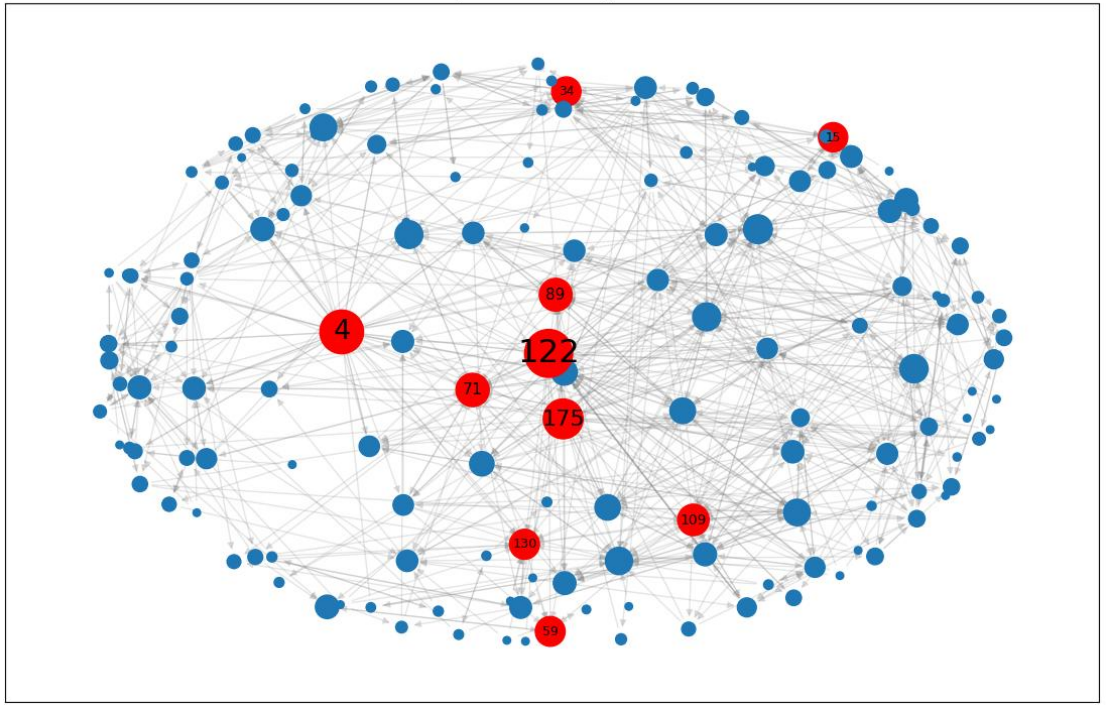


After applying PageRank, I found that the output results have **changed significantly** compared to before. When using different degrees, only one **node [4]** is the same in the obtained results. This indicates that this method is more capable of **highlighting** the difference in the degree of **attention and being attended** to.

Top 10 Authorities with PageRank



Top 10 Hubs with PageRank



We can observe that the common top 10 for all the results is the node [4]. This indicates that it has extensive and significant influence, being the **most popular** node in the network and playing a **crucial role in the connection** of the network structure.

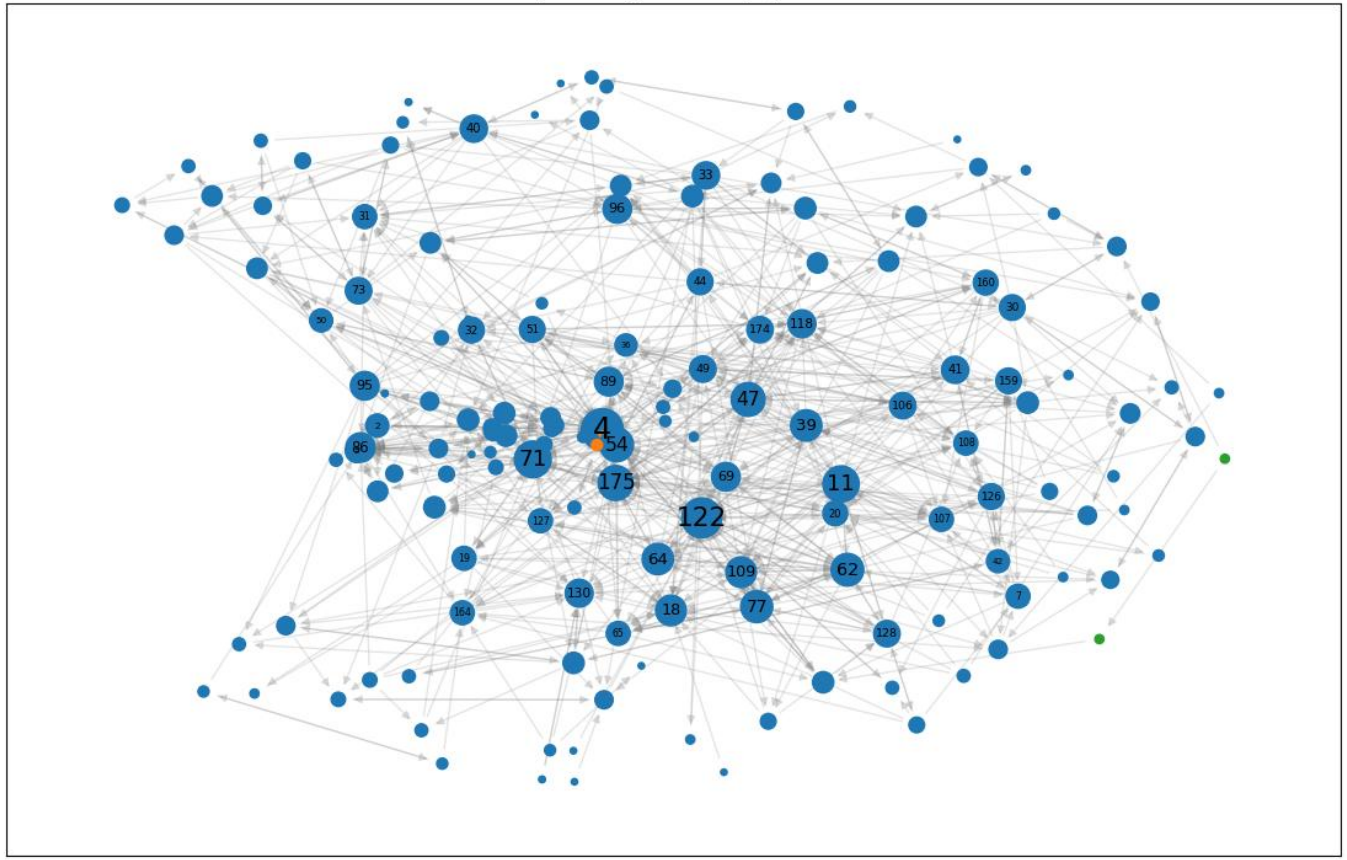
I was very curious about whom the node is , so I output its ID to recheck the detailed information. The analysis of the results revealed that it is the **most renowned game producer** in China in recent years.

I use three methods to find communities:

Label Propagation method has identified 3 communities [152, 2, 2], but the first community almost encompasses all the nodes, while the remaining two communities each have **less than three nodes**.

This indicates that the algorithm **is not suitable** for detecting the communities in my network. The modularity is: 0.006138.

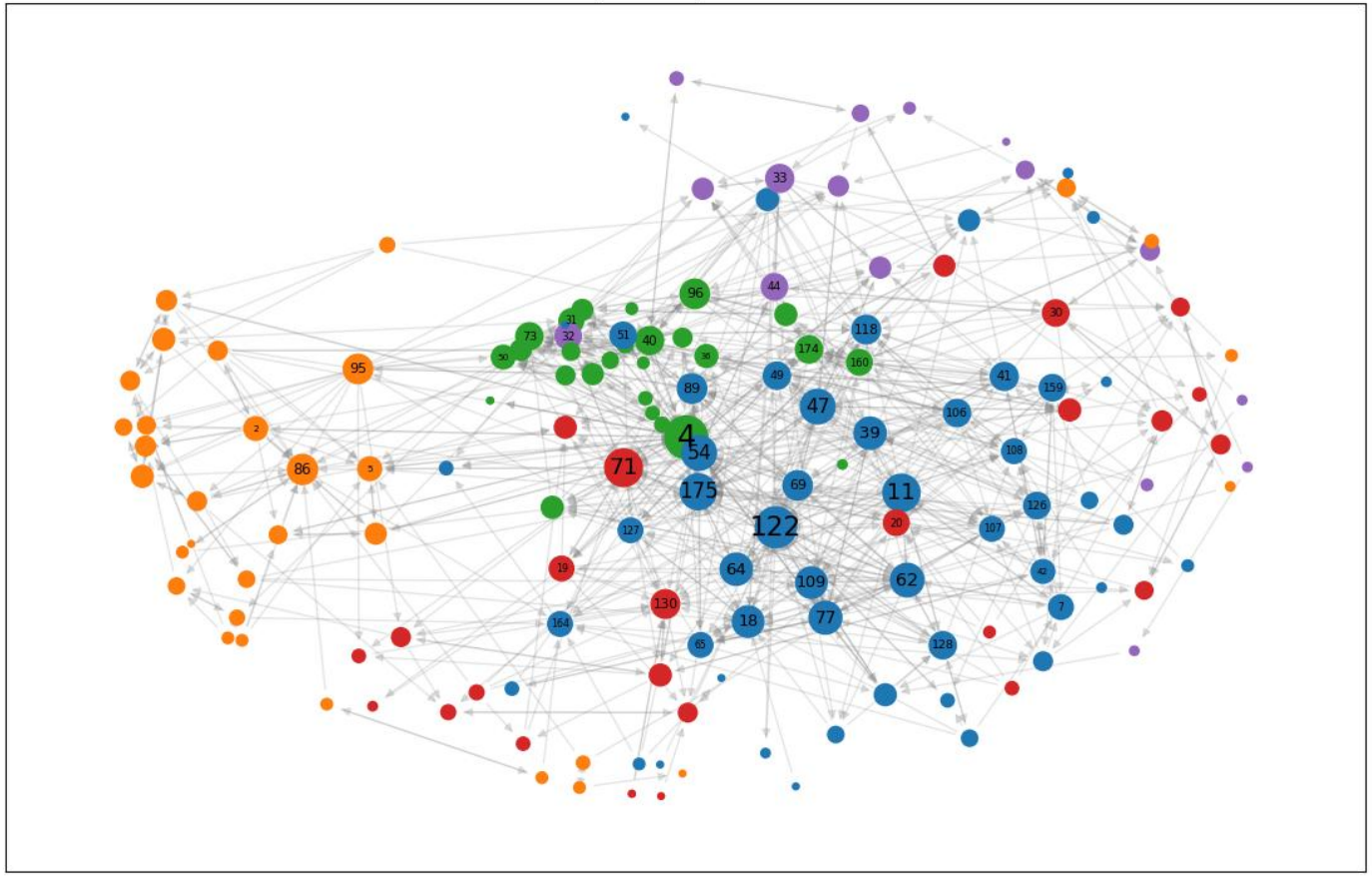
Community found by Label Propagation method



The Louvain method identified 5 communities. They have **nearly the same number** of members. Moreover, it is clearly visible in the image that different communities were distributed in relatively concentrated **different areas**. This indicates that the communities discovered by this algorithm are **more reasonable**.

The modularity is: 0.3909.

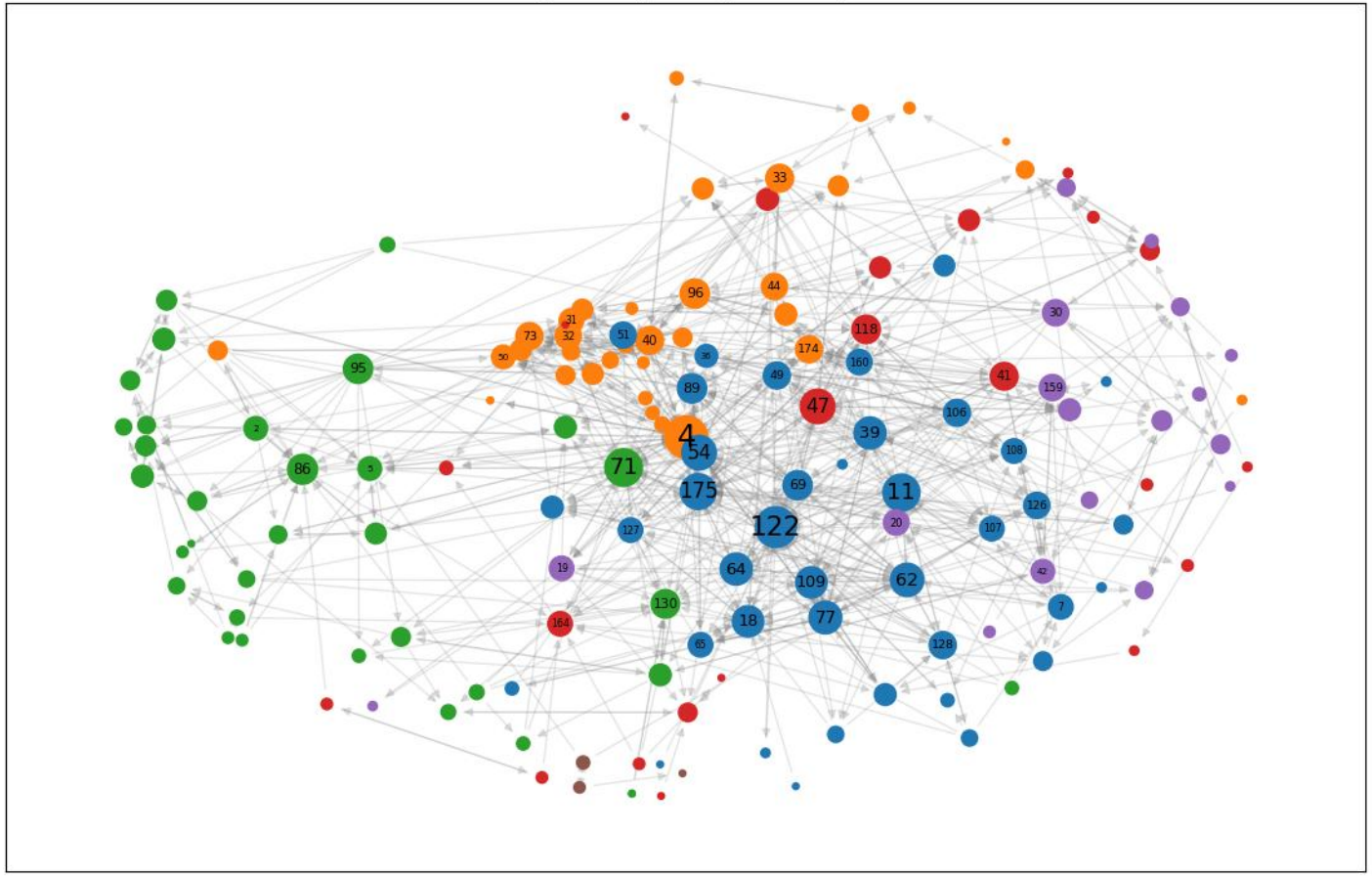
Community found by Louvain method



The Greedy Modularity method identified 6 communities **[39, 38, 34, 24, 18, 3].** The number of members in the first five communities is very **close to** that found by **Louvain**. However, the **last community** has only three members, which is very small, so I think it **should be excluded**.

The modularity is: 0.3753.

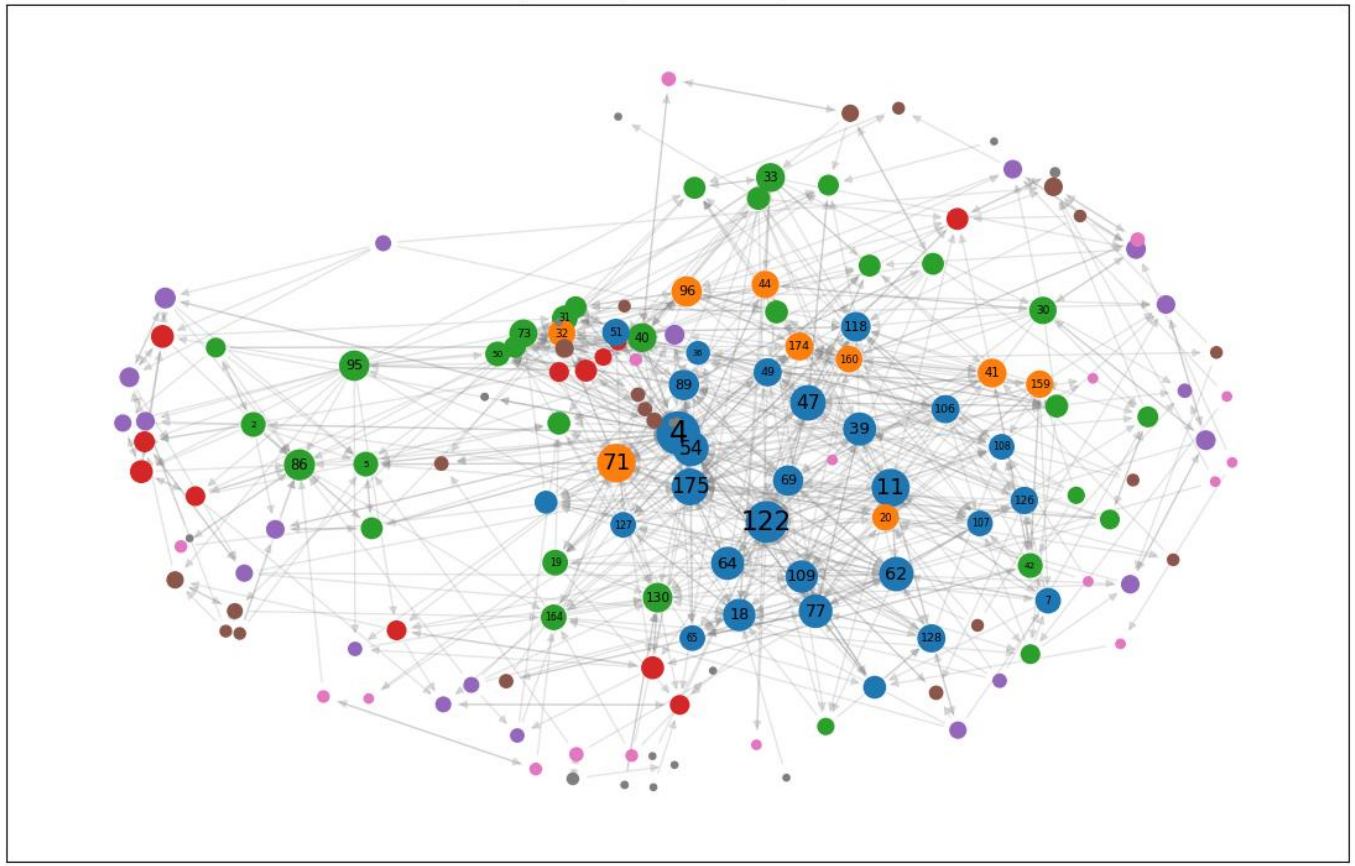
Community found by Greedy Modularity method



The k-core decomposition results of the network show that the **maximum core** number is **8**, and the distribution of the number of nodes at each level is **[28, 9, 31, 12, 21, 20, 17, 18].** The modularity calculation result is **0.1593**. The number of **6-core** nodes is the **largest**, making them the main force of the network core; the **8-core** is the **innermost** core layer. The modularity is relatively low.

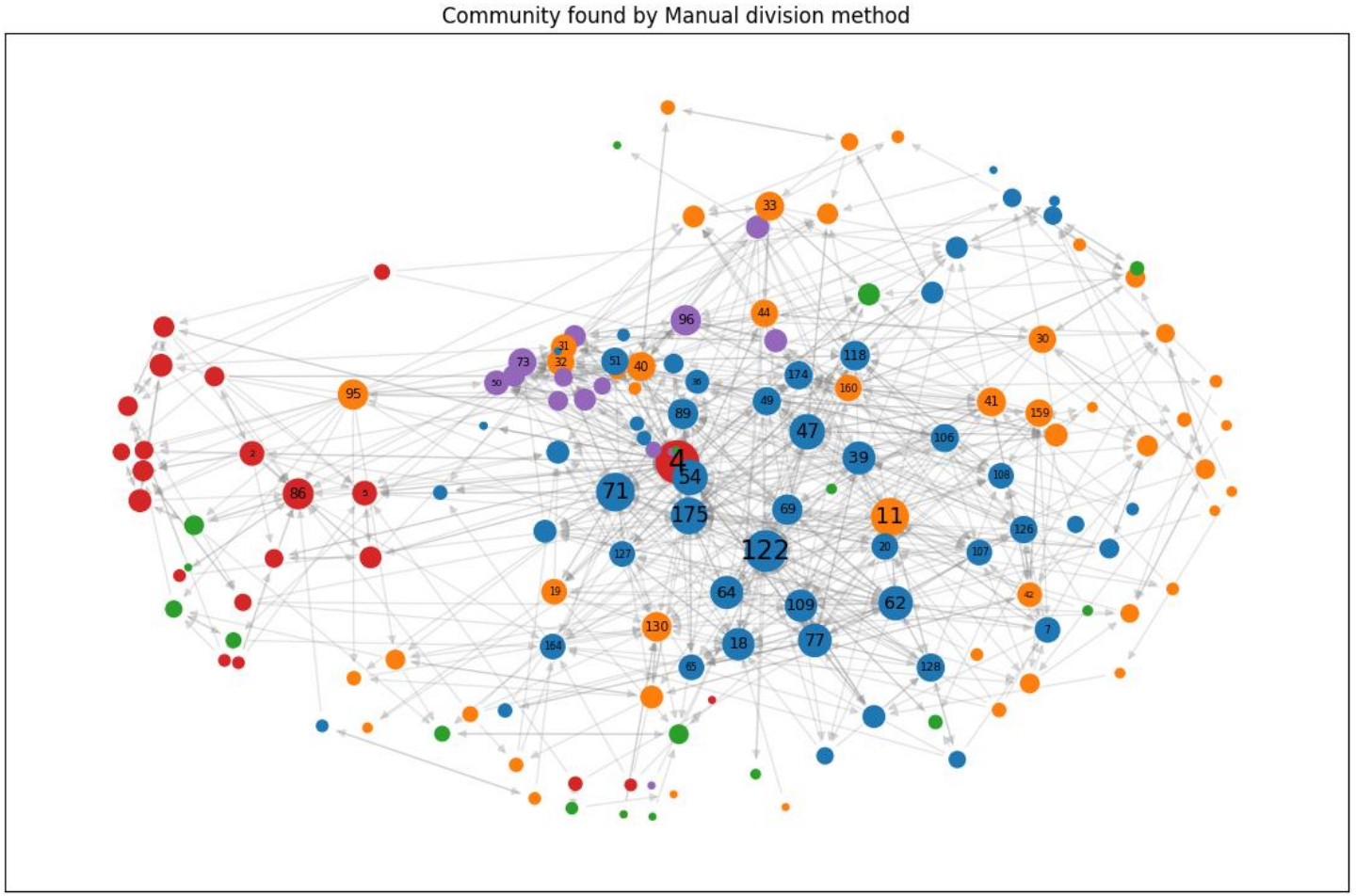
K-core color order:
8: blue
7:orange
6 :green
5 :red
4 :purple
3 :brown
2 :pink
1 :gray

Community found by K-core decomposition method



After manual classification, 5 communities were obtained **[39, 38, 30, 28, 21],** each representing the divisions where the people I follow belong. The specific details are as follows: **{'film': 15, 'game': 22, 'other': 17, 'science': 49, 'society': 53}.**

The modularity is: 0.20



1. I constructed a directed and unweighted network with 156 nodes and 761 edges through the following relationship on Weibo.

2. Through analysis, I confirmed that it conforms to the scale-free property, the small-world property and is disassortative.

3. By comparing it with random networks, it was found that it is most similar to the Barabasi Albert model.

4. Based on my analysis, among the top 10 nodes in the network, [4] is the most popular.

5. Through community algorithms, it was found that the number of communities obtained by the Louvain and Greedy Modularity algorithms was in good agreement with the manual division.